



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Situation: Review the scenario below. Then complete the step-by-step instructions.

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets

- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

Summary	Recently, it was reported to the security team that employees were unable to access internal network resources. Network logs indicate that the server received a large number of ICMP packets, eventually causing it to overload and reject incoming connections. After analyzing the logs, the team found that the ICMP packets had been sent through an unconfigured firewall, which was allowing unwanted traffic into the network
Identify	The incident team audited the network and server logs to identify the security gaps. Through this, the team found an unconfigured firewall was allowing unwanted traffic into the network. The attacker was able to leverage this firewall and, through it, employ an ICMP flood attack.
Protect	The team has since implemented and improved security measures. Starting by updating the baseline configurations of the firewalls to prevent unwanted traffic and limit the rate of incoming ICMP packets. Secondly, an IP verification system was implemented to prevent spoofed IPs on incoming ICMP packets. The team also implemented IDS and IPS to monitor the network for known attack patterns and signatures and act upon recognition.
Detect	To detect incoming attacks, the team has opted to use an IDS to monitor and alert of malicious ICMP traffic based on suspicious patterns and signatures
Respond	The team blocked all incoming ICMP packets and took all non-critical services offline. This allowed the team to isolate the attack and prevent any other services from being affected. The team also updated the base configurations

	of firewalls to ensure all are filtering and monitoring incoming ICMP traffic for known malicious patterns or traces. The team also used an IPS to act upon IDS alerts
Recover	Once the Firewall was configured, IDS was installed and running, and the malicious ICMP packets were correctly filtered out, and access to internal network services was restored to internal network users

Reflections/Notes: